Practical Incentive-Compatible Drafts: Carry-Over Lottery Allocation

Paper Track: Basketball

Open-source repository: https://github.com/kvr06-ai/cola-sloan-ssac27

Introduction

For decades, the NBA has navigated the tradeoff between helping weak teams via the draft and the tanking that ensues. Carry-Over Lottery Allocation (COLA) solves the dilemma. COLA defines weakness using multi-season playoff results, not single-season records. Thus, tanking need not be rewarded. Draft priority carries over between seasons, being reduced when teams have playoff success or make high draft picks. Over time, more help is directed to weaker teams.

COLA is a framework where each mechanism may have different rules on how draft priority changes. COLA's definition includes the NBA's own 3-2-1 lottery as an edge case: minimal carry-over exclusions and no reward for losing. COLA principles could also inform credit allocation in auction-based drafts.

We prove that, under reasonable assumptions, COLA can be incentive-compatible while helping the weakest team, circumventing Munro and Banchio's SSAC 2020 impossibility result for draft mechanisms based on end-of-season records. Only one assumption requires justification: teams prefer winning games over manipulating the pool of lottery teams. Others are standard assumptions.

Methods

The incentive-compatibility proof is conventional; novelty comes from the reframe that makes it possible.

We discuss tradeoffs within the COLA framework by identifying tunable dials:

- How does a team's draft priority increase?
- How does the priority decrease?
- How does priority map to lottery odds and draft positions?

ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move of realistic fictional leagues, lets us compare different draft mechanisms using these criteria:

- **Targeted help.** How much help do the highest priority teams receive?
- **Long-term parity.** Can all teams be competitive, as measured by playoff success?
- **Incentive compatibility.** Does the mechanism reward losing?

Simplicity is assessed qualitatively. In the paper, we analyze the impact of violating model assumptions (e.g., a team preferring the lottery over the playoffs) and how to mitigate issues. Generally, mitigation increases complexity.

Results

We ran nine mechanisms on the same 30 simulated leagues for 25 seasons each and compared them by paired differences with 95% confidence intervals.

The old NBA lottery rewards losing, which caused tanking. 3-2-1 removes that reward but does not help weak teams. Simple COLA also eliminates tanking, and it directs the most help to teams that have waited longest (Table 1). All three are level on parity. On simplicity, Simple COLA wins outright: no lottery is used. Teams are ordered by most years without a playoff series win or a top-3 pick; ties are broken by most wins this season. It is one example of many COLA mechanisms that solve the core dilemma.

	Incentive compatibility	**Targeted help**		**Long-term parity**
Rule	**Help for a worse record** *places gained, 5th-worst to worst*	**Help for a longer wait** *places gained, 5th-longest to longest*	**Longest wait for a series win or top-3 pick** *seasons*	**Wait for the next series win** *seasons, average*
Old NBA lottery	$+1.32$	0	16.4	4.65
New lottery, 3-2-1	-0.63	0	16.3	4.47
Simple COLA	-0.94	6.3	5.5	4.58

Table 1. Three mechanisms on the same 30 simulated leagues of 25 seasons (means over leagues). Help counts the draft places one team gains when only its record, or only its wait, changes, so a positive number is an earlier pick. The full paper provides a fuller analysis, including rewards for losing from other starting positions and simulation results for six additional mechanisms.

Conclusion

COLA shows that helping the weakest teams and ending tanking are not mutually exclusive. It only looked that way because "weakest" was measured by end-of-season record. Simple COLA resolves the dilemma without complexity, and the wider family lets a league choose its own balance of targeted help, parity, simplicity, and robustness.